

MCMINNVILLE, OR, USA

WMO: 726881

Lat: 45.195N

Lon: 123.134W

Elev: 49

StdP: 100.74

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 94273

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.4	-2.1	-10.7	1.5	-1.1	-7.0	2.1	1.5	12.0	9.8	10.9	9.4	1.7	350	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.0	33.8	19.4	31.6	19.0	29.1	18.2	20.5	30.9	19.6	29.8	18.7	28.2	3.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.0	23.2	15.9	11.4	22.5	15.0	10.7	21.6	58.9	30.7	55.6	29.9	52.9	28.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.8	6.8	-6.7	37.9	2.9	1.9	-8.7	39.2	-10.4	40.3	-12.0	41.3	-14.1	42.7		
(5)			-7.3	23.1	2.8	1.8	-9.3	24.4	-11.0	25.5	-12.6	26.5	-14.6	27.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	5.3	6.4	8.1	10.1	13.5	16.4	20.0	20.1	17.2	12.1	7.7	4.9	(6)	(7)
	DBStd	6.16	3.28	2.85	2.66	2.99	3.29	3.19	2.78	2.76	3.18	3.00	3.29	3.51	(7)	(8)
	HDD10.0	620	148	104	71	33	6	0	0	0	0	13	83	162	(8)	(9)
	HDD18.3	2544	404	334	316	247	153	76	13	13	58	195	318	416	(9)	(10)
	CDD10.0	1296	2	3	13	37	116	191	309	313	217	78	14	4	(10)	(11)
	CDD18.3	179	0	0	0	1	4	18	63	67	24	1	0	0	(11)	(12)
	CDH23.3	2807	0	0	0	21	119	322	966	935	423	21	0	0	(12)	(13)
	CDH26.7	1151	0	0	0	3	32	125	423	408	157	3	0	0	(13)	(14)
(14) Wind	WSAvg	2.6	2.6	2.8	2.9	2.7	2.4	2.5	2.6	2.4	2.2	2.3	2.6	2.7	(14)	(15)
(15) Precipitation	PrecAvg	1088	175	127	118	71	51	26	11	16	39	80	161	194	(15)	(16)
	PrecMax	1965	347	335	213	178	110	68	56	118	192	184	322	508	(16)	(17)
	PrecMin	618	9	9	21	8	0	0	0	0	0	6	16	5	(17)	(18)
	PrecStd	252	92	78	49	41	28	19	13	23	39	46	81	106	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.0	20.8	26.2	30.3	34.2	37.1	36.3	33.9	26.2	17.8	16.5	(19)	(20)
		MCWB	12.3	10.3	13.4	16.5	18.5	19.8	20.8	19.8	18.9	14.9	14.8	13.2	(20)	(21)
	2%	DB	12.5	13.8	17.4	22.3	26.9	30.2	33.7	33.6	31.0	22.4	15.7	13.1	(21)	(22)
		MCWB	10.6	9.7	11.9	14.2	17.1	18.6	19.5	19.4	17.9	14.2	13.1	11.4	(22)	(23)
	5%	DB	11.2	12.3	15.0	18.7	23.6	27.1	31.2	31.3	28.0	19.9	14.0	11.2	(23)	(24)
		MCWB	9.6	9.3	10.7	12.4	15.7	17.4	19.2	18.9	17.1	13.4	11.6	9.7	(24)	(25)
	10%	DB	10.0	11.1	13.0	16.2	21.0	24.0	28.7	28.6	25.6	17.6	12.7	9.6	(25)	(26)
		MCWB	8.5	8.6	9.9	11.2	14.6	16.1	18.4	18.3	16.3	13.2	10.7	8.1	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.2	14.4	17.4	19.8	21.1	22.4	21.5	20.6	17.7	15.7	13.5	(27)	(28)
		MCDB	14.4	13.8	18.7	25.3	28.4	31.7	34.2	32.7	30.0	22.5	17.2	15.8	(28)	(29)
	2%	WB	11.1	10.9	12.7	14.9	17.8	19.1	20.5	20.2	19.0	15.8	13.8	11.7	(29)	(30)
		MCDB	12.0	12.7	15.8	21.0	25.0	28.7	31.0	30.9	28.0	19.6	15.3	13.0	(30)	(31)
	5%	WB	9.9	9.8	11.5	13.0	16.3	17.8	19.5	19.3	17.9	14.7	12.2	10.0	(31)	(32)
		MCDB	11.0	11.5	14.3	17.6	22.5	26.0	30.1	29.5	25.8	18.0	13.4	11.1	(32)	(33)
	10%	WB	8.7	8.7	10.3	11.7	14.9	16.6	18.5	18.5	16.9	13.8	11.1	8.2	(33)	(34)
		MCDB	9.7	10.6	12.9	15.3	20.3	23.4	28.2	27.9	23.8	16.8	12.4	9.2	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.6	9.0	10.6	12.0	12.9	16.0	16.0	14.7	10.5	7.2	5.7	(35)	(36)
		MCDBR	7.0	9.6	12.9	16.5	17.9	18.8	20.6	20.7	19.8	15.4	8.4	6.6	(36)	(37)
	5% WB	MCWBR	5.3	6.5	8.1	9.6	9.7	8.8	8.4	8.3	8.8	8.5	5.8	5.1	(37)	(38)
		MCWBR	6.3	7.7	11.2	14.3	15.9	16.8	19.0	18.4	17.3	11.1	6.4	6.1	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.322	0.319	0.331	0.359	0.370	0.362	0.352	0.355	0.345	0.334	0.327	0.328	(39)	(40)
		taud	2.518	2.535	2.504	2.392	2.378	2.411	2.460	2.467	2.488	2.527	2.510	2.463	(40)	(41)
	Ebn at Noon		794	872	905	902	898	904	909	895	878	839	778	739	(41)	(42)
		Edh at Noon	67	78	92	112	117	114	107	103	93	78	67	65	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		1.21	2.04	2.97	4.31	5.49	6.08	6.94	5.93	4.43	2.58	1.40	0.98	(43)	(44)
	RadStd		0.19	0.28	0.39	0.39	0.54	0.56	0.46	0.39	0.38	0.31	0.19	0.15	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	+26

Nomenclature: See separate page